

TKG Test ROM Manual

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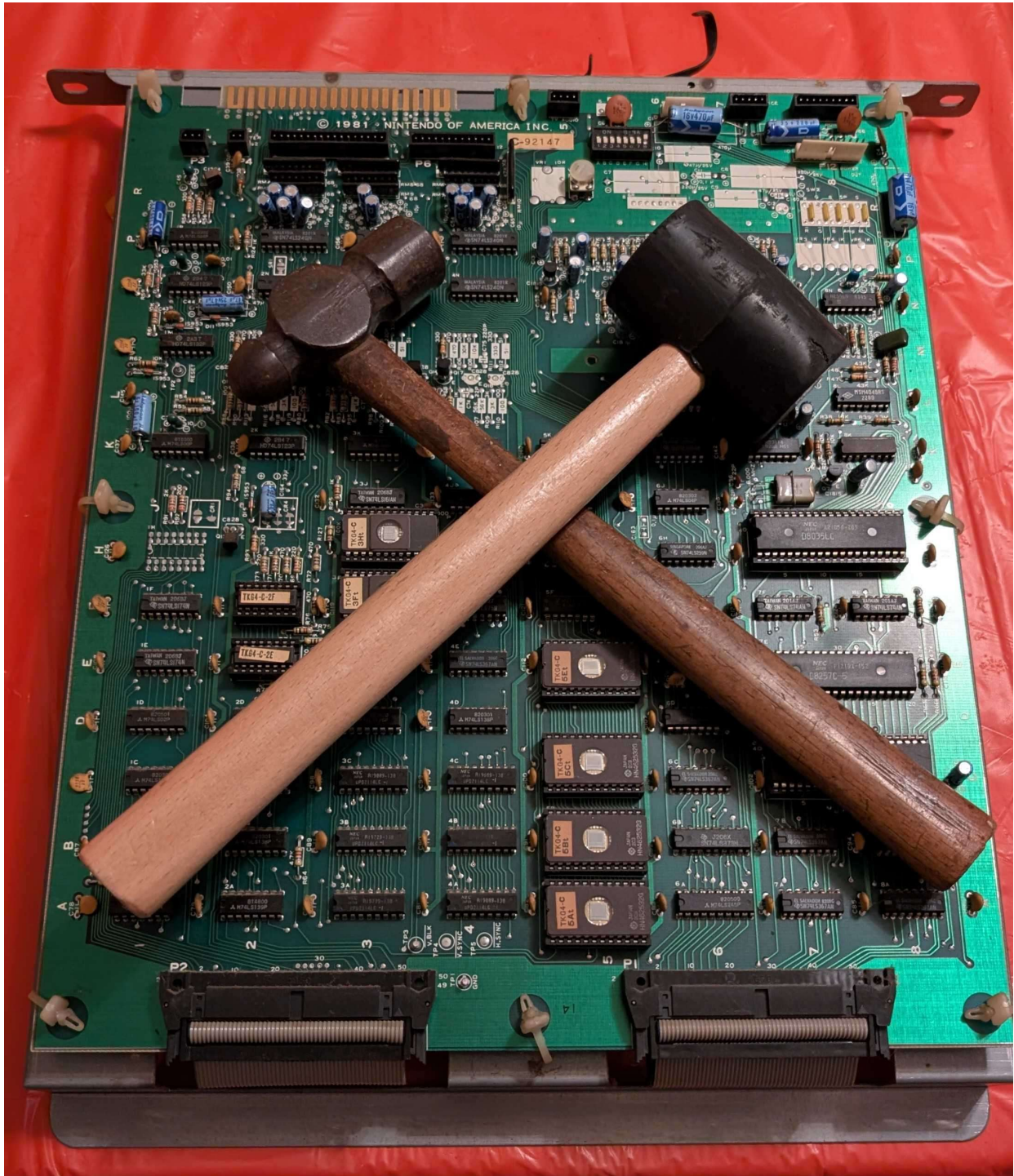


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Chapter 1: Overview

Requirements

This test ROM will not work under every condition. Some things need to be taken into consideration:

- All boards involved must have power, in this case +5V, +12V, and -5V must all be present and within appropriate tolerance, usually within +/- 5%.
- Clocks must be present enough to run. The CPU runs at approximately 3MHz.
- The reset pin and NMI pins must not be held in active. If both are always active, then the ROM will not boot.
- The CPU must be able to access address 0x0000 to 0x0FFF. If the game does not boot it cannot access these ROMs.
- Address and data buffers must be working, if they are not it will not be possible for the program to run.
- Audio OR video must be working. If neither are working, the test ROM will have no way to report failures
- The ROM socket for 5F or 5E (depending on which hardware being tested) must be in good enough condition to make proper contact.

Compatible Hardware

This test ROM is intended for original TKG hardware. It is compatible with the following hardware:

- Fully compatible
 - TKG2
 - TKG3
 - TKG4
- Partially compatible
 - TRS2 – RAM, ROM, DMA, and NMI tests should all work. However, it is a different game so audio tests will not work. Monitor tests might also not work correctly.

This test ROM will work in the cocktail, cabaret, and upright models of the machine.

Using the Test ROM

Using the test ROM is very simple. The ROM is burnt onto a 2532 4k EPROM. In order to run the tests you place the EPROM into location 5F for TKG2/TKG3 hardware and 5E TKG4 hardware. Note the notch in the socket, EPROM, and pin 1 markers. After the ROM is inserted reconnect the board back to the cabinet or bench setup for testing.

There are two groups of tests performed by the test ROM: Startup and Runtime. Startup tests are tests that are automatically run when the game boots. This tests RAM, ROM, DMA, NMI, and Audio tests. All tests with the exception of audio tests must pass in order to transition from startup to runtime. Runtime tests are additional tests used for diagnosing other issues with the machine in question. It contains things such as a monitor test and control printouts.

TKG START-UP TESTS V0.15 22 AUG 2026				TKG RUN-TIME TESTS V0.15 22 AUG 2026			
RAM TESTS				CONTROLS TEST			
RAM 0L	PASS	RAM 3L	PASS	P1 RIGHT	OFF	P2 RIGHT	OFF
RAM 0H	PASS	RAM 3H	PASS	P1 LEFT	OFF	P2 LEFT	OFF
RAM 1L	PASS	RAM 4L	PASS	P1 UP	OFF	P2 UP	OFF
RAM 1H	PASS	RAM 4H	PASS	P1 DOWN	OFF	P2 DOWN	OFF
RAM 2L	PASS			P1 JUMP	OFF	P2 JUMP	OFF
RAM 2H	PASS	B. TEST	PASS	P1 START	OFF	P2 START	OFF
				COIN	OFF		
ROM CHECKSUMS				DIP SW TEST			
ROM 0	F31E			00000001			
ROM 1	73BA			ABCDEFGH			
ROM 2	B2BC						
ROM 3	A0F0						
SYSTEM TESTS				SELECT A TEST			
DMA	PASS			? PALETTE TEST			1
NMI	PASS			SCREEN INVERT			0
				MONITOR TEST			0
				SOUND			0
				MUSIC			0
				BACKGROUND TEST			
				SPRITE TEST			
				RESET			
AUDIO TESTS							
SOUND	6						
MUSIC	4						

Chapter 2: Startup Tests

RAM Tests

RAM Tests

RAM Bank Test

TRS2 and TKG2 RAM Locations

Name	Designation	Type	Start Address	Size	High Bit Location	Low Bit Location
RAM0	Work RAM	2114	0x6000	1k x 4	C-6L	C-6H
RAM1	Work RAM	2114	0x6400	1k x 4	C-6M	C-6J
RAM2	Work RAM	2114	0x6800	1k x 4	C-6N	C-6K
RAM3	Sprite	2148	0x7000	1k x 4	V-5J	V-5K
RAM4	Video/Background	2114	0x7400	1k x 4	V-3K	V-3J

TKG3 RAM Locations

Name	Designation	Type	Start Address	Size	High Bit Location	Low Bit Location
RAM0	Work RAM	2114	0x6000	1k x 4	C-6L	C-6H
RAM1	Work RAM	2114	0x6400	1k x 4	C-6M	C-6J
RAM2	Work RAM	2114	0x6800	1k x 4	C-6N	C-6K
RAM3	Sprite	2148	0x7000	1k x 4	V-2L	V-2M
RAM4	Video/Background	2114	0x7400	1k x 4	V-5L	V-5M

TKG4 RAM Locations

Name	Designation	Type	Start Address	Size	High Bit Location	Low Bit Location
RAM0	Work RAM	2114	0x6000	1k x 4	C-4C	C-3C
RAM1	Work RAM	2114	0x6400	1k x 4	C-4B	C-3B
RAM2	Work RAM	2114	0x6800	1k x 4	C-4A	C-3A
RAM3	Sprite	2148	0x7000	1k x 4	V-6R	V-6P
RAM4	Video/Background	2114	0x7400	1k x 4	V-2R	V-2P

ROM Checksums

Test Overview

The ROM test performs a 16-bit checksum on the contents of all ROMs. This test does not report the integrity of the ROMs nor does it validate that it is correct for a location. That is up to the user to determine. The exception to this is ROM 0, which is where the test ROM is located.

Test ROM Integrity Test

As part of the ROM test, ROM 0 will be validated. ROM 0 is the test ROM itself. Its own checksum is appended to the footer as part of a post-build script. It will compare the measured checksum with what is expected. If there is any corruption of the test ROM or if there is issues with accessing say the upper addresses, it will report a failure by printing “Integrity Fail”. In this case, the test ROM may not be trustworthy and could run into issues during execution.



ROM Locations

ROM ID	ROM Type	ROM Size	Start Address	TKG2/TKG3 Location	TKG4 Location
ROM 0	2532	4k	0x0000	5F	5E
ROM 1	2532	4k	0x1000	5G	5C
ROM 2	2532	4k	0x2000	5H	5B
ROM 3	2532	4k	0x3000	5K	5A

ROM Checksum Table

ROM ID	US Set 1	US Set 1 Hard	US Set 2	Japan Set 1	Japan Set 2	Japan Set 3
ROM 0						
ROM 1	73BA					
ROM 2	B2BC					
ROM 3	A0F0					

DMA Test

NMI Test

Audio Tests

Sound Test

Music Test

Transition to Runtime

Chapter 3: Runtime Tests

Controls Test

DIP Switch Test

Palette Test

Screen Invert

Monitor Test

Sound Test

Music Test

Background Test

Sprite Test

Reset

Chapter 4: Test Architectures

Abbreviations

CPU: Central Processing Unit

NMI: Non-Maskable Interrupt

RAM: Random Access Memory

ROM: Read Only Memory